Faculty of Science, Engineering and Computing

Module: CI5210 Networking Concepts Setter: Martin Tunnicliffe
Title of Assignment: IP Address Scheme Design and Static Routing Student: **Mohamud Elmi**
Deadline: **08 January** 202**4** Module weighting: 30%

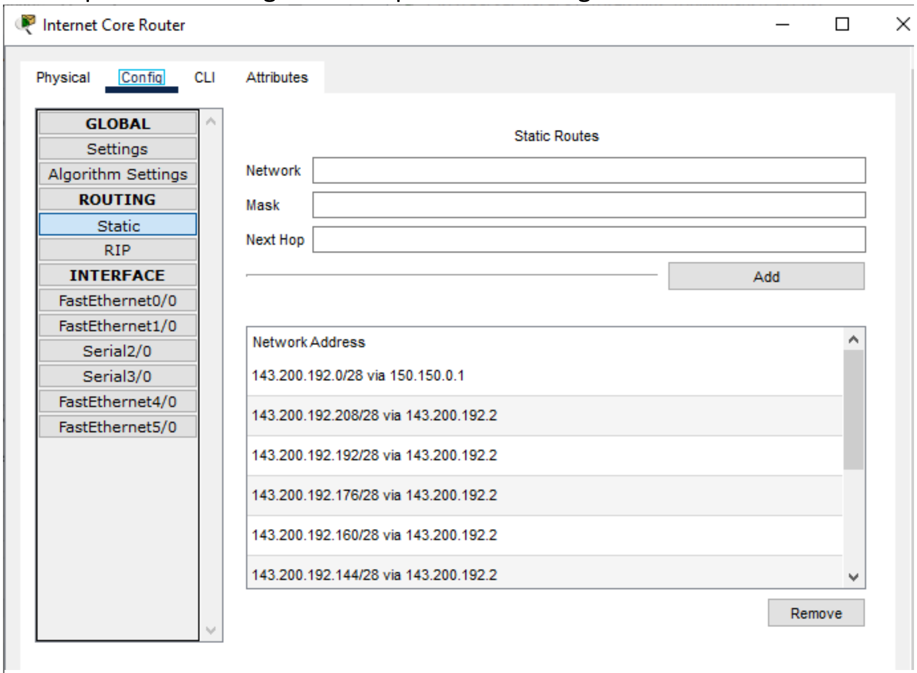
Introduction:

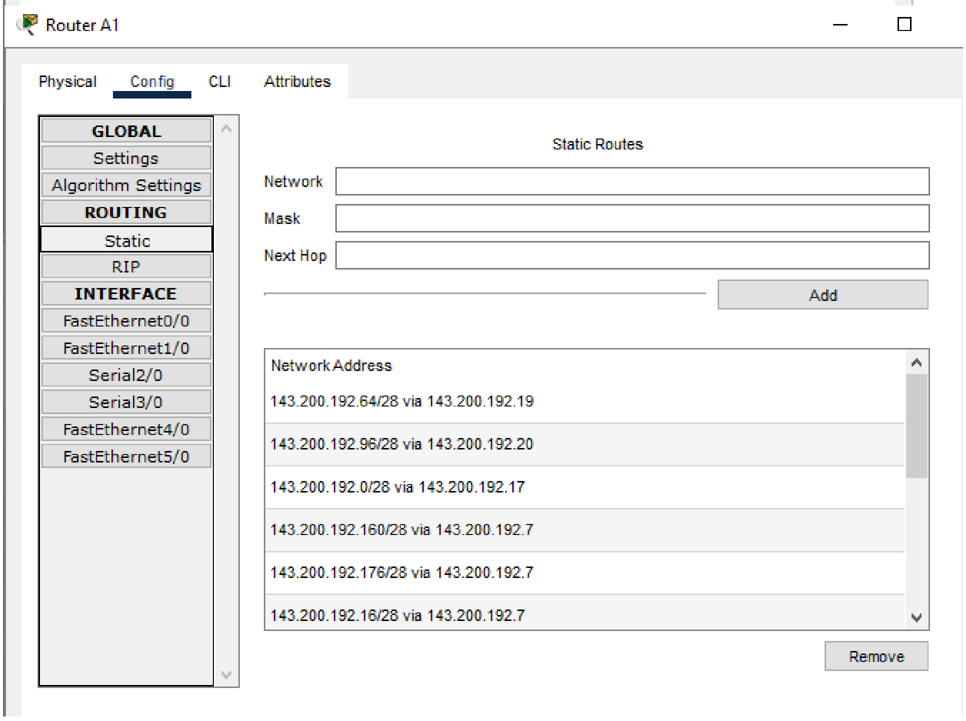
For this project, we will develop an addressing plan utilizing Variable Length Subnet Masking (VLSM) for a small two-location network with sites in Kingston and Surbiton.
The network topology depicting router connections is provided in Diagram 1.
Our addressing scheme will efficiently allocate IP addresses across subnetworks of varied sizes within the intranet.
Additionally, we will set up static routing protocol specifications on the routers to facilitate communication between the distinct subnetworks as well as with the external Internet.
This will demonstrate proficiency in strategic IP allocation and routing configuration for an interconnected multi-subnet system.

IP Addressing Scheme:
For the domain address allocated to the network (143.200.192.0/24), we will use a VLSM addressing scheme with a supernet mask of 255.255.255.0. This mask provides enough host addresses for the number of hosts in each subnet while minimizing the number of subnets required.
The Kingston site consists of two buildings, each with three workgroup networks and a backbone network. The Subiton site has a single building with two workgroup networks and a backbone network. We will assign the following IP addresses and masks to each subnet and backbone device:

Device	Interface	IP Address	Subnet Mask
Kingston Router A1	FA0/0	143.200.192.33	255.255.255.240
Kingston Router A1	FA1/0	143.200.192.18	255.255.255.240
Kingston Router A2	FA0/0	143.200.192.65	255.255.255.240
Kingston Router A2	FA1/0	143.200.192.19	255.255.255.240
Kingston Router A3	FA0/0	143.200.192.97	255.255.255.240
Kingston Router A3	FA1/0	143.200.192.20	255.255.255.240
Kingston Router B1	FA0/0	143.200.192.145	255.255.255.240
Kingston Router B1	FA1/0	143.200.192.114	255.255.255.240
Kingston Router B2	FA0/0	143.200.192.161	255.255.255.240
Kingston Router B2	FA1/0	143.200.192.115	255.255.255.240
Kingston Router B3	FA0/0	143.200.192.177	255.255.255.240
Kingston Router B3	FA1/0	143.200.192.116	255.255.255.240
WAN Router 1	FA0/0	143.200.192.5	255.255.255.240
WAN Router 1	SE2/0	143.200.192.21	255.255.255.240
WAN Router 2	SE2/0	143.200.192.22	255.255.255.240
WAN Router 2	FA0/0	143.200.192.4	255.255.255.240
Domain Gateway Router	FA0/0	143.200.192.2	255.255.255.240
Domain Gateway Router	FA1/0	150.150.0.1	255.255.0.0
Internet Core Router	FA1/0	150.150.0.2	255.255.0.0
Internet Core Router	FA0/0	150.250.0.1	255.255.0.0
Internet Server	FA0	150.250.127.56	255.255.0.0

Static Routing Scheme:
We will design an addressing plan for a small network with two locations - Kingston and Surbiton. We will use Variable Length Subnet Masking (VLSM) to make better use of IP addresses.
The Kingston location has two buildings. Each building has three department networks and one backbone router. The Surbiton location has one building. This building has two department networks and one backbone router.
We will carefully choose IP addresses and subnet masks for each network device. This will give enough addresses for the needed hosts in each subnet. It will use IP addresses wisely across the whole network.
The assigned addresses and masks aim to meet current and future needs for each location, building and department. Using VLSM helps avoid wasting IP addresses in this interconnected network.





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Testing and Evaluation:
Using the routing setup and packet tracer's simulation, we can test if devices can connect in a safe environment. We will use ping tests and traceroute maps. These show if hosts on different department networks can reach each other. They also show the path data takes. We will check if all hosts can get to web pages on the Internet Server. Checking step-by-step that connections work as expected this way helps make sure things will run smoothly when launched for real use.

Pinging from Lab A1 to Lab A3

```
Pinging 143.200.192.146 with 32 bytes of data:

Reply from 143.200.192.146: bytes=32 time=22ms TTL=124
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Reply from 143.200.192.146: bytes=32 time=22ms TTL=124

Ping statistics for 143.200.192.146:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 22ms, Maximum = 22ms, Average = 22ms

C:\>
```

Pinging from Lab A3 to B1

```
C:\>tracert 143.200.192.162

Tracing route to 143.200.192.162 over a maximum of 30 hops:

  1   4 ms      4 ms      4 ms      143.200.192.65
  2   117 ms    0 ms      0 ms      143.200.192.17
  3   0 ms      0 ms      1 ms      143.200.192.6
  4   0 ms      0 ms      0 ms      143.200.192.115
  5   34 ms     1 ms      0 ms      143.200.192.162

Trace complete.

C:\>
```

Tracert from Lab A2 to B2

```
C:\>ping 143.200.192.34

Pinging 143.200.192.34 with 32 bytes of data:

Reply from 143.200.192.34: bytes=32 time=2ms TTL=123
Reply from 143.200.192.34: bytes=32 time=1ms TTL=123
Reply from 143.200.192.34: bytes=32 time=82ms TTL=123
Reply from 143.200.192.34: bytes=32 time=127ms TTL=123

Ping statistics for 143.200.192.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 127ms, Average = 53ms

C:\>
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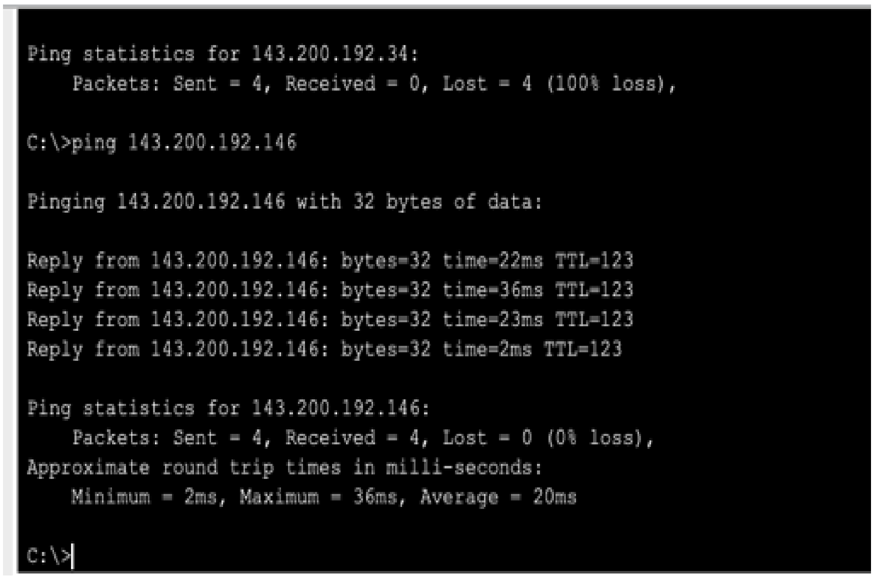
Pinging fromLab A1 to Office C1



```
Ping statistics for 143.200.192.34:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 143.200.192.146

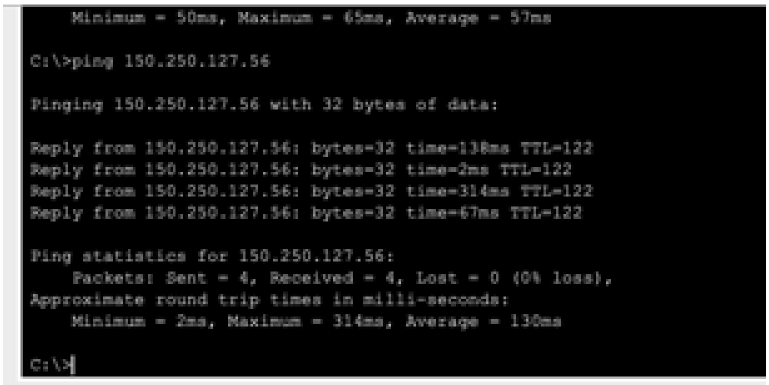
Pinging 143.200.192.146 with 32 bytes of data:

Reply from 143.200.192.146: bytes=32 time=22ms TTL=123
Reply from 143.200.192.146: bytes=32 time=36ms TTL=123
Reply from 143.200.192.146: bytes=32 time=23ms TTL=123
Reply from 143.200.192.146: bytes=32 time=2ms TTL=123

Ping statistics for 143.200.192.146:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 36ms, Average = 20ms

C:\>
```

Pinging Lab B3 to Office C2



```
    Minimum = 50ms, Maximum = 65ms, Average = 57ms

C:\>ping 150.250.127.56

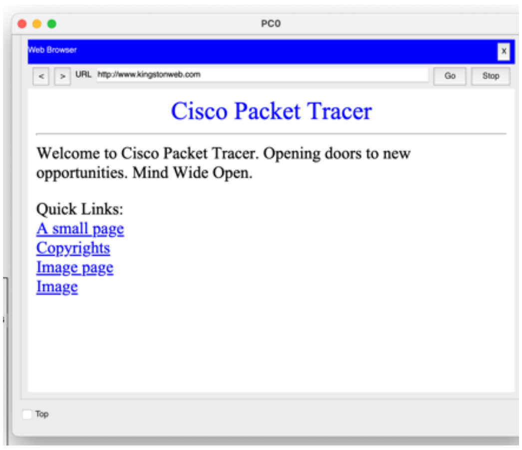
Pinging 150.250.127.56 with 32 bytes of data:

Reply from 150.250.127.56: bytes=32 time=138ms TTL=122
Reply from 150.250.127.56: bytes=32 time=2ms TTL=122
Reply from 150.250.127.56: bytes=32 time=314ms TTL=122
Reply from 150.250.127.56: bytes=32 time=67ms TTL=122

Ping statistics for 150.250.127.56:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 314ms, Average = 130ms

C:\>
```

Pinging Lab A1 to Internet Server



The screenshot shows a workstation accessing the Internet Server's website. For this to work across devices, we set up the Internet Server as a DNS server. This translates names into IP addresses.
We gave the Internet Server a domain name (like www.example.com) mapped to its IP address. Client devices can now convert this name to its IP address when making requests.
We also added the Internet Server's DNS details to each workstation's settings. This lets workstations translate the domain name to the right IP address for routing requests.
With DNS resolution configured network-wide, requests use hostnames that get converted to target IP addresses behind the scenes. This enables seamless access to the Internet Server web page across the infrastructure.

Conclusion:
We designed an IP addressing plan for a small two-location network using Variable Length Subnet Masking (VLSM). The sites are Kingston and Surbiton.
Routers at each site were assigned static routes, enabling them to build routing tables. Routers ask neighbours about routes, constructing a map to all destinations.
The network uses the address block 143.200.192.0/24. This consolidates addressing while allowing subnet sizes to fit each site's needs. One segment connects to the Internet.
Careful addressing and static routing connect all devices across sites.